

Debanjan Pal

Senior Game Developer | VR & XR Specialist

4.5+ Years Exp • Unity 2D/3D • Meta Quest VR • C# Architecture • Motion Sensors

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SCAN PORTFOLIO

PROFESSIONAL SUMMARY

Results-driven **Senior Game Developer** with **4.5+ years of professional industry experience** architecting high-performance 2D/3D games, virtual reality medical simulations, and custom hardware motion-control systems. Highly proficient in Unity (URP, Physics, Cinemachine), clean decoupled C# architecture, and Meta Quest 2/3/3S deployment. Recognized for rapid gameplay prototyping, optimizing memory/draw calls, and delivering immersive interactive applications from concept to production.

PROFESSIONAL EXPERIENCE

Senior Game Developer — Equidor Medtech

April 2024 – Present

- Lead gameplay programming for cross-platform 2D/3D titles and interactive VR experiences, driving core loops from whitebox to release.
- Architect specialized applications interfacing directly with proprietary motion-based sensor hardware and custom controllers.
- Champion clean engineering workflows, sprint planning, peer code reviews, profiling, and memory optimization across devices.

Associate Software Developer — Cyclops Medtech

Jan 2022 – March 2024 (2 yrs 3 mos)

- Engineered, optimized, and shipped medical-grade diagnostic VR applications targeting Meta Quest 2 (Android/XR Toolkit).
- Analyzed telemetry data, diagnostic sensor logs, and system crash dumps to identify bottlenecks and maintain high frame rates.
- Collaborated closely with product owners and clinicians to translate complex clinical protocols into intuitive, gamified patient tests.

TECHNICAL SKILLS & ARSENAL

GAME DEV & ENGINES

Unity (2D/3D), Universal Render Pipeline (URP), 2D/3D Physics, Cinemachine, UI Toolkit & Canvas, Mecanim Animation, Particle Systems.

VR & HARDWARE SYSTEMS

Meta Quest 2 / 3 / 3S, XR Interaction Toolkit, Motion Sensors, Spatial 6DoF Tracking, Custom Controllers, Medical-Grade VR, Telemetry.

LANGUAGES & ARCHITECTURE

C# (.NET), Design Patterns, State Machines (FSM), Object Pooling, Observer/Events, Data Structures, Decoupled Architecture.

TOOLS & 3D CREATION

Blender 3D, Photoshop, Git / GitLab / GitHub, Unity Profiler, Memory Optimization, Draw Call Batching, JIRA & Agile.

FEATURED PRODUCTION & PORTFOLIO PROJECTS

Walk & Dodge — 3D Obstacle Course Platformer

Unity 3D • C# • Physics • Cinemachine

- Engineered 3D obstacle avoidance game with dynamic hazards, moving traps, responsive character physics, and dynamic multi-angle tracking.
- Implemented **Object Pooling** for obstacle instantiation, locking 60 FPS gameplay with zero memory allocation spikes.

Bird Runner — Procedural Endless Runner

Unity • C# • Procedural Generation • Mobile

- Built an infinite runner loop with dynamic procedural obstacle spawning, incremental speed curves, and high-score persistence.
- Optimized draw calls and lightweight collision matrices specifically tailored for smooth frame rates on mobile hardware.

Survival Center — 3D Resource & AI Simulation

Unity 3D • C# • NavMesh AI • State Machine

- Architected modular state-driven entity behaviors using **NavMesh Pathfinding** for autonomous agent navigation and resource loops.

EDUCATION & CREDENTIALS

BCA in Game and Mobile App Development

CMR University, Bangalore

2019 – 2022

VR Development Fundamentals in Unity

Udemy Professional Credential

2022

High School Diploma

Siliguri Boys' High School

2011 – 2019